

Name: _____

Period: _____

Date: _____

Topic 1.6 Guided Notes: Hearing and the Other Senses (2 of 2)

Unit 1: Biological Bases of Behavior | AP Psychology

Directions: Read through the topic overview page below and fill in each blank as you go. This worksheet covers Parts 3-5 (Hearing, Chemical Senses, and Body Senses) — Thresholds and Vision are on the first worksheet for this topic.

Source page: appsychreviewhub.com/overviews/topic1.6

Part 3: Hearing (Audition)

Word Bank

<i>Conduction Deafness</i>	<i>Middle Ear</i>	<i>Outer Ear</i>	<i>Inner Ear</i>
<i>Cochlea</i>	<i>Sound Localization</i>	<i>wavelength</i>	<i>amplitude</i>
<i>Hair Cells</i>	<i>Sensorineural Deafness</i>	<i>Place Theory</i>	<i>Frequency Theory</i>
<i>Volley Theory</i>			

Hearing involves the transduction of sound waves (changes in air pressure) into neural signals. A sound wave's _____ determines the pitch we hear, while its _____ determines the loudness.

The Three Sections of the Ear

- _____: the visible part (pinna) that funnels sound waves down the auditory canal to the eardrum, causing it to vibrate.
- _____: the chamber between the eardrum and cochlea containing three tiny bones (hammer, anvil, and stirrup) that concentrate the vibrations of the eardrum onto the cochlea's oval window.
- _____: the innermost part of the ear, containing the _____ (a coiled, bony, fluid-filled tube where sound waves trigger nerve impulses), the semicircular canals, and vestibular sacs. _____ embedded in the cochlea's basilar membrane bend when the fluid moves, actually transducing sound into electrical signals.

Theories of Pitch Perception

- _____: links the pitch we hear with the exact place where the cochlea's membrane is stimulated. Best explains high pitches.
- _____: the rate of nerve impulses traveling up the auditory nerve matches the frequency of a tone, enabling us to sense its pitch. Best explains low pitches.
- _____: groups ("volleys") of neurons take turns firing in rapid succession, combining their firing rates to keep up with tones too high for any single neuron to track alone.

_____ is how we identify where a sound is coming from, mostly by comparing tiny differences in timing and intensity as a sound reaches each ear separately.

- _____: occurs when damage to the mechanical structures that conduct sound to the cochlea (like the eardrum or middle ear bones) prevents vibrations from getting through.
- _____: usually more serious and permanent, resulting from damage to the cochlea's hair cells or the auditory nerve itself.

Apply It: Which Hearing Concept?

#	Scenario	Concept
1	A boxer takes a hard hit to the side of the head, rupturing his eardrum; sound vibrations can no longer properly reach his cochlea, though the cochlea itself is undamaged.	
2	A very low foghorn causes a listener's auditory neurons to fire in a rhythm that matches the slow rate of the sound wave itself.	
3	Even with her eyes closed, a woman immediately points to the exact spot in the room where a phone is ringing, based on tiny differences in when the sound reaches each of her ears.	

Part 4: The Chemical Senses (Taste and Smell)

Word Bank

<i>Nontasters</i>	<i>Smell</i>	<i>Medium Tasters</i>	<i>Pheromones</i>
<i>Supertasters</i>	<i>Oleogustus</i>	<i>Taste</i>	

Unlike vision and hearing, which respond to energy waves, taste and smell respond to chemical molecules.

- _____ (Gustation): a chemical sense utilizing receptors in taste buds located on the papillae of the tongue. The basic tastes are sweet, sour, salty, bitter, umami (savory), and the more recently recognized _____, the taste of fat.
- _____: people with a higher-than-average number of taste buds, especially sensitive to bitter flavors.
- _____: people with an average number of taste buds.
- _____: people with relatively few taste buds, who need much stronger flavors to notice the same tastes.
- _____ (Olfaction): olfactory receptors in the nasal cavity catch airborne molecules. Unlike all other senses, smell bypasses the thalamus and goes directly to the olfactory bulb, which is closely connected to the limbic system — this is why smells often trigger vivid emotional memories. Smell also detects _____, chemical signals released by an organism that can communicate information to, and influence the behavior of, other members of the same species.

Part 5: The Body Senses (Touch, Pain, Movement, and Balance)

Word Bank

<i>Gate Control Theory</i>	<i>Vestibular Sense</i>	<i>Touch</i>
<i>Kinesthetic Sense</i>	<i>Phantom Limb Syndrome</i>	

_____ (Somatosensation) relies on a network of receptors in the skin and other tissues that detect pressure, temperature, and pain. Interestingly, the sensation of "hot" isn't produced by a single dedicated heat receptor — it results from your warm and cold receptors firing at the same time, which your brain interprets as "hot."

Psychologists still don't have one single, complete explanation for how pain works. The most influential attempt is _____, which proposes that the spinal cord contains a neurological "gate" that can either block pain signals from reaching the brain or let them through, depending on other sensory input — part of why rubbing an injury can genuinely make it hurt less.

_____ occurs when a person who has lost a limb continues to feel sensation, including pain, seemingly coming from the limb that's no longer there.

- _____: the system for sensing the position and movement of individual body parts, with receptors located in your muscles, joints, and tendons.
- _____: the sense of body movement and position, including balance, relying on fluid in the semicircular canals and vestibular sacs in the inner ear.

Apply It: Which Body Sense or Pain Theory?

#	Scenario	Concept
1	Years after losing his leg in an accident, a veteran still occasionally feels an intense itching sensation in his missing foot.	
2	A gymnast performing a routine with her eyes closed can still feel exactly how her body is tilted and rotating in the air.	
3	A doctor lightly touches both a warm probe and a cool probe to a patient's arm at the exact same spot simultaneously, and the patient reports feeling an intense burning heat.	

Video Notes: Homunculus (Crash Course Psychology #6)

This video covers hearing, taste, smell, touch, and body senses together — answer the questions below as you watch; some ideas will already be familiar from Parts 3-5, and that's fine.

- According to the video, what are the three tiny bones in the middle ear, and what is their job?
- What are the five basic taste sensations mentioned in the video?
- According to the video, why does a particular smell often bring back a specific memory or emotion?

- What are the four basic touch sensations described in the video?
- According to the video, what is kinesthesia, and what does the vestibular sense monitor?

6. Don't Trip Up! (Common Misconceptions)

_____ is about your knees, joints, and individual body parts. _____ is about your overall head/body balance and is controlled by the inner ear.

7. Quick Check: Review Questions

Circle the best answer for each question below.

1. With her eyes completely closed, a dancer can still accurately sense the exact position and movement of her arms, legs, and fingers as she performs a routine. This ability relies primarily on her:
 - A. Vestibular Sense
 - B. Olfaction
 - C. Gustation
 - D. Kinesthetic Sense
2. Years of exposure to loud concerts permanently damages the delicate hair cells inside a musician's cochlea, resulting in significant, irreversible hearing loss even though his eardrum and middle ear bones are completely intact. This type of hearing loss is called:
 - A. Sensorineural Deafness
 - B. Somatosensation Loss
 - C. Gate Control Failure
 - D. Conduction Deafness
3. A sound engineer determines that a very high-pitched tone stimulates a highly specific location toward the beginning of the cochlea's membrane. Which theory of pitch perception best explains how this high pitch is being detected?
 - A. Frequency Theory
 - B. Volley Theory
 - C. Opponent-Process Theory
 - D. Place Theory

4. After stubbing her toe, a woman instinctively rubs the area and reports that doing so genuinely makes the pain feel less intense. Which theory of pain best explains why rubbing an injury can reduce the perceived pain?

- A. Gate Control Theory
- B. Weber's Law
- C. Opponent-Process Theory
- D. Place Theory

5. Unlike every other sense, which travels through the thalamus before reaching higher brain regions, this particular sense bypasses the thalamus entirely and goes directly to the olfactory bulb, which connects closely to the limbic system, explaining why it triggers such vivid emotional memories. This sense is:

- A. Taste (Gustation)
- B. Smell (Olfaction)
- C. Hearing (Audition)
- D. Vision



Check Your Progress

Done with your notes? Head over to the [Quiz](https://appspsychreviewhub.com/quiz) to test what you've learned:

<https://appspsychreviewhub.com/quiz>